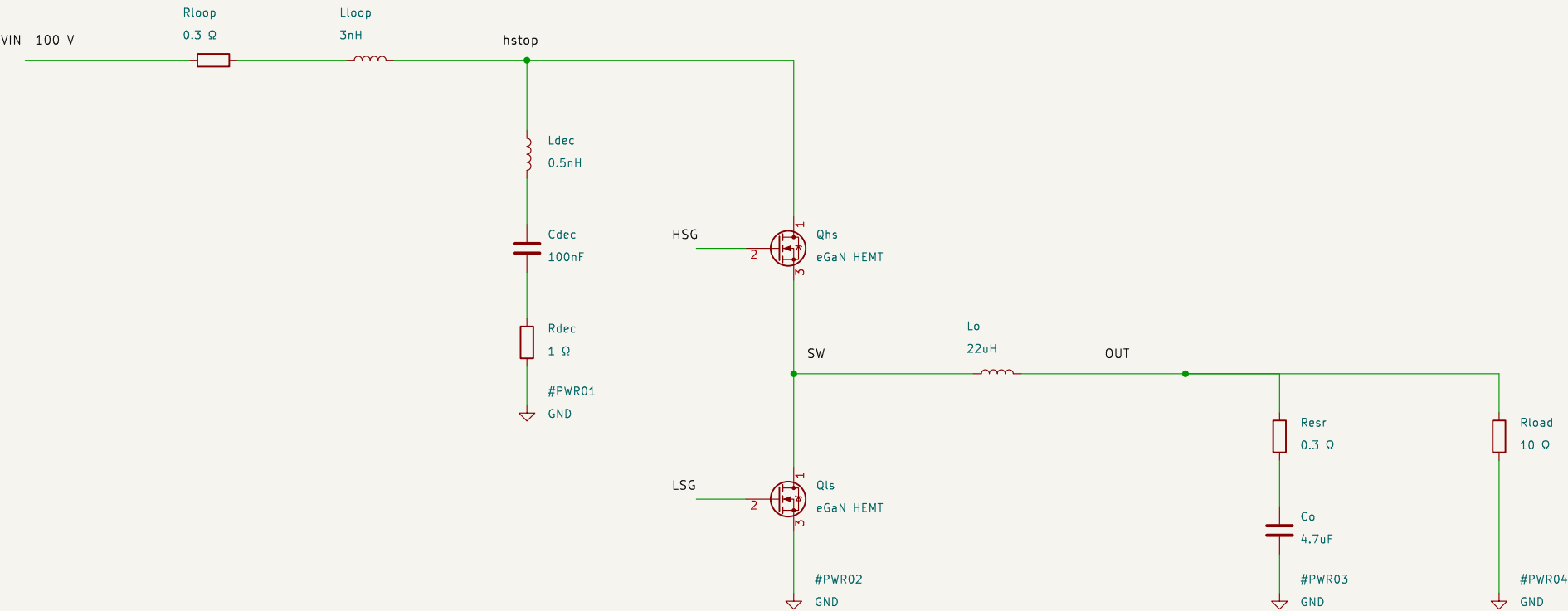


GaN synchronous buck converter – the circuit simulated in sim/buck.cir
100 V in, 50.0 V out at 500 kHz. Values read from the netlist at build time.



Qhs / Qls are enhancement-mode GaN HEMTs. KiCad has no GaN symbol, so an n-channel enhancement MOSFET symbol is used, as in EPC's own schematics. The device has NO BODY DIODE – reverse conduction during dead-time costs $V_{th} + |V_{off}| + I \cdot R_{ds(on)}$, not one diode drop. This sets the dead-time trade.

Ldec / Cdec / Rdec is the bus decoupling. Rdec = 1 ohm is a damping value, not a parasitic: at 20 mOhm this branch rings at ~22 MHz and drove the low-side gate to +11.8 V against a 1.4 V threshold at a 200 V bus.

HSG / LSG come from the segmented gate driver (8 pull-up + 8 pull-down slices, active Miller clamp, -2 V off rail) driven by seg_gate_ctrl.v.

Not drawn: Vsin, Vshs, Vsls, Vsout are 0 V sources in the netlist used only to sense current. They are measurement points, not components.